Lucid: End-to-End Application Methodology

Integrated Three-Workflow Lifecycle: Portfolio Creation, Demat Diagnosis & Continuous Drift Monitoring

WORKFLOW I: NEW PORTFOLIO CREATION METHODOLOGY (STAGES 4.1 – 4.8)

ONBOARDING → FUNNEL → AI VETTING → SLSQP → WHOLE SHARES

1. Onboarding (Stage 4.1)

- **Capital: ≥ ₹10,000**
- Horizon: 3 to 360 Months
- 5-Question Risk Score: Conservative / Moderate / Agg.
- NLP Sector Exclusions

Output: Structured Mandate

2. Regime & Funnel (4.2-4.4)

- **VIX Regime Detection:** Calm (<13) | Mod (14.2) | Vol | Ext
- **3-Tier Liquidity Funnel:** 2,639 Master → 504 Liquid Pool
- Factor Z-Scores: Quality, Val, Mom

Output: 30-60 Candidate Pool

3. AI Committee & Math (4.5-4.7)

- **LangGraph Multi-Agent Team:** Macro • Market • Fund • News • Critic
- **SciPy SLSQP Solver (Max Sharpe):** Stock ≤25% • Sector ≤40%
- **Whole-Share Integer Allocation:**

Output: Qi Whole Lots & Cash ≥ ₹0

4. Portfolio & Why (Stage 4.8)

- **Discrete Whole Holdings:** e.g. HDFCBANK 32 sh, RELIANCE 16
- **Key Performance Metrics:** Div Score 95.5 • Sharpe 0.41
- 2-Level Stock-Why Explanations

Committed Active Portfolio

Track Active Portfolio Every 300s

WORKFLOW II: DEMAT DIAGNOSIS (4.9)

CAS AUDIT

1. CAS Statement Ingestion

CDSL / NSDL PDF or Broker CSV Upload

pdfplumber 0.11.5 Statement Engine

2. 3-Step Entity Resolution

Matches across 2,639 listed stocks

ISIN Match → pg_trgm (>0.4) → Levenshtein

3. HHI Concentration Audit

HHI = ∑ (wi · 100)²

<1500 Safe | 1500-2500 Mod | >2500 High Risk

4. 7-Section Diagnostic Report

Flags Single-Stock (>25%) & Sector (>40%)

Interactive Copilot Chat: /diagnosis/{id}/chat

Suggests Remediation Swaps

Bridge → Rebalance into Disciplined Portfolio

Rebalan

WORKFLOW III: 300s DRIFT RADAR & REBALANCING

STAGES 4.10 – 4.12

Stage 4.10: Continuous 300s Drift Radar Daemon

Runs Mon–Fri 09:15–15:30 IST • Session Advisory Lock: pg_try_advisory_lock(847291)

Live Revaluation against NSE Quotes • Freshness Guards: LIVE / 15-MIN DELAYED / EOD

Drift Breach?
Stock ≥5% | Sec ≥8%

No Breach → **Branch A: Hold**
Status: ON_TRACK

Breach Detected

Stage 4.11: Before / After Rebalance Preview

BEFORE (Drifted State)
IT 34.5% (+6.5% Drift) → SELL 4 Sh

→ **AFTER (Realignment Target v+1)**
IT 28.0% (Aligned) • FMCG 17.5% (+8 Sh)

Stage 4.12: Statutory Indian Cost Engine & Order Execution

Mandatory Levies (October 2024 Schedule):
STT (0.10%) • NSE/BSE (0.003%) • Stamp (0.015%) • GST (18%)

Cash Guarantee
C_res ≥ ₹0 (Trims 1 Sh)